

Worksheet on differentiation - MEMO.

Note Title

2014/03/20

$$1. \frac{dy}{dx} = 3x^2 + 3^x \cdot \ln 3$$

$$2. \frac{dy}{dx} = 3(\cos(4^{5x}))^2 \cdot (-\sin(4^{5x})) \cdot 4^{5x} \cdot \ln 4 \cdot 5$$

$$3. 3x^2y^4 + 4x^3y^3 \cdot y' + 5x^4 - 6y^5 \cdot y' = 0$$

$$\Rightarrow y' [4x^3y^3 - 6y^5] = -3x^2y^4 - 5x^4$$

$$\Rightarrow y' = \frac{-3x^2y^4 - 5x^4}{4x^3y^3 - 6y^5}$$

$$4. f'(x) = \cos(e^{-2x}) \cdot e^{-2x} \cdot -2$$

$$5. f'(t) = e^{\cos \sqrt{t}} (-\sin \sqrt{t}) \frac{1}{2\sqrt{t}}$$

$$6. f'(t) = \frac{-\frac{1}{t^2}}{\sqrt{1 - (\frac{1}{t})^2}}$$

$$7. f'(x) = \frac{12x^3}{1 + (3x^4)^2}$$

$$8. f'(v) = \frac{-3}{1 - 3v}$$

$$9. f'(x) = \frac{2}{2x} = \frac{1}{x}$$

$$10. f'(t) = \frac{\frac{1}{t}}{\ln t}$$

$$11. f'(x) = 0$$

$$12. f'(t) = -\operatorname{cosec}(\cot(9t)) \cot(\cot(9t)) (-\operatorname{cosec}^2(9t)) 9$$

$$13. f'(x) = 10(x-x^3)^9 \cdot (1-3x^2)$$

$$14. f'(t) = 10^{1-2t} \cdot \ln 10 \cdot (-2)$$

$$15. f(x) = (1-2x)^{10x}$$

$$\Rightarrow \ln f(x) = 10x \ln(1-2x)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = 10 \ln(1-2x) + 10x \cdot \frac{-2}{1-2x}$$

$$\Rightarrow f'(x) = f(x) \left[10 \ln(1-2x) - \frac{20x}{1-2x} \right]$$

$$16. \ln f(x) = \ln(1-x^3) - \ln(7 + 4 \arcsin(5x))$$

$$\frac{f'(x)}{f(x)} = \frac{-3x^2}{1-x^3} - \frac{\frac{4 \cdot 5}{\sqrt{1-(5x)^2}}}{7 + 4 \arcsin(5x)}$$

$$f'(x) = \dots \dots \dots \text{(very ugly)}$$

$$17. f'(x) = \frac{-1}{1+(1-x)^2} \cdot \cos kx + \arctan(1-x) \cdot (-\sin kx) \cdot k$$

$$18. f(x) = 2x + \ln \tan 2x - 2 \ln \sec(x^2)$$

$$f'(x) = 2 + \frac{2 \sec^2 2x}{\tan 2x} - \frac{2 \cdot \sec(x^2) \tan(x^2) \cdot 2x}{\sec(x^2)}$$

$$19. f'(x) = 2 \sin 2x \cdot \cos 2x \cdot 2$$

$$f'\left(\frac{\pi}{6}\right) = 4 \sin \frac{\pi}{3} \cdot \cos \frac{\pi}{3} = 4\left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right) = \sqrt{3}$$

If $x = \frac{\pi}{6}$ then $f(\frac{\pi}{6}) = \sin^2(\frac{\pi}{3}) = (\frac{\sqrt{3}}{2})^2$
 $= \frac{3}{4}$

Tangent has slope $\sqrt{3}$ through $(\frac{\pi}{6}, \frac{3}{4})$

$$\Rightarrow y - \frac{3}{4} = \sqrt{3}(x - \frac{\pi}{6})$$

$$y = \sqrt{3}x - \frac{\sqrt{3}\pi}{6} + \frac{3}{4}$$

20. $9x^2 + 4y^2 = 36$

$$\Rightarrow 18x + 8y \cdot y' = 0$$

$$y' = \frac{-18x}{8y} = -\frac{9x}{4y}$$

Parallel to $y = 2x \Rightarrow y' = -\frac{9x}{4y} = 2$
 $\Rightarrow y = -\frac{9}{8}x$

Say the tangent touches at $x = a$
 $\Rightarrow y = -\frac{9}{8}a$ so that $(a, -\frac{9}{8}a)$ lies
 on the ellips.

Therefore $9(a)^2 + 4(-\frac{9}{8}a)^2 = 36$

$$9a^2 + 4 \cdot \frac{81}{64} a^2 = 36$$

$$a^2(9 + \frac{81}{16}) = 36$$

$$a^2(\frac{225}{16}) = 36$$

$$a^2 = \frac{36 \cdot 16}{225}$$

$$a = \pm \frac{(6)(4)}{15} = \pm 1\frac{3}{5}$$

